Screening for fetal well-being in a high-risk pregnant population comparing the nonstress test with umbilical artery Doppler velocimetry: A randomized controlled clinical trial

Keith P. Williams, MBBS,[a] Duncan F. Farquharson, MD,[b] Michael Bebbington, MD,[c] Jerome Dansereau, MD,[d] France Galerneau, MD,[a] R. D. Wilson, MD,[e] Dorothy Shaw, MD,[b] and Nancy Kent, MD[b]

New Haven, Conn, Vancouver, British Columbia, Canada, Bronx, NY, and Philadelphia, Pa

OBJECTIVE: The purpose of this study was to evaluate the ability of two different modes of antepartum fetal testing to screen for the presence of peripartum morbidity, as measured by the cesarean delivery rate for fetal distress in labor.

STUDY DESIGN: Over a 36-month period, all patients who were referred to the Fetal Assessment Unit at BC Women's Hospital because of a perceived increased fetal antepartum risk at a gestational age of ≥32 weeks of gestation were approached to participate in this study. Fetal surveillance of these women was allocated randomly to either umbilical artery Doppler ultrasound testing or nonstress testing as a screening test for fetal well-being. If either the umbilical artery Doppler testing or the nonstress testing was normal, patients were screened subsequently with the same technique, according to study protocol. When the Doppler study showed a systolic/diastolic ratio of >90th percentile or the nonstress testing was equivocal (ie, variable decelerations), an amniotic fluid index was performed, as an additional screening test. When the amniotic fluid index was abnormal (<5th percentile), induction and delivery were recommended. When the Doppler study showed absent or reversed diastolic blood flow or when the nonstress test result was abnormal, induction and delivery were recommended to the attending physician. Statistical comparisons between groups were performed with an unpaired *t* test for normally distributed continuous variables and χ^2 test for categoric variables.

RESULTS: One thousand three hundred sixty patients were assigned randomly to groups in the study; 16 patients were lost to follow up. Six hundred forty-nine patients received Doppler testing and 691 received nonstress testing. The mean number of visits for the Doppler test and nonstress test groups was two versus two, respectively. The major indications for fetal assessment included postdates (43%), decreased fetal movement (22%), diabetes mellitus (11%), hypertension (10%), and intrauterine growth restriction (7%). The incidence of cesarean delivery for fetal distress was significantly lower in the Doppler group compared with the nonstress testing group (30 [4.6%] vs 60 [8.7%], respectively; $P < .006$). The greatest impact on the reduction in cesarean deliveries for fetal distress was seen in the subgroups in which the indication for testing was hypertension and suspected intrauterine growth restriction.

CONCLUSION: Umbilical artery Doppler as a screening test for fetal well-being in a high-risk population was associated with a decreased incidence of cesarean delivery for fetal distress compared to the nonstress testing, with no increase in neonatal morbidity. (Am J Obstet Gynecol 2003;188:1366-71.)

Key words: Umbilical artery Doppler, nonstress test, antepartum fetal monitoring

From the Department of Obstetrics and Gynecology, Yale University School of Medicine,[a] the Diagnostic and Ambulatory Program, BC Women's Hospital, University of British Columbia,[b] the Departments of Obstetrics and Gynecology and Women's Health, Montefiore Medical Center, Albert Einstein College of Medicine.[c] Victoria General Hospital,[d] and the Center for Fetal Diagnosis and Treatment, Children's Hospital of Philadelphia.[e]
Supported in part by the British Columbia Medical Services Foundation, which had no involvement in any aspects of the study.
Received for publication July 9, 2002; revised September 18, 2002; accepted January 23, 2003.
Reprint requests: Keith P. Williams, MD, 333 Cedar St/PO Box 208063, New Haven, CT 06520-8063. E-mail: keith.williams@yale.edu

0002-9378/2003 $30.00 + 0
doi:10.1067/mob.2003.305

The evaluation of the antenatal fetal heart rate (FHR) pattern with electronic fetal monitoring is a widely accepted screening test of fetal well-being. Nonstress tests (NSTs) are used in an attempt to reduce the incidence of fetal compromise at birth that is the result of placental insufficiency.[1] Fetal compromise has been associated with changes in baseline FHR, repetitive decelerations, and prolonged decreased variability of the FHR.[2] As a screening test, however, FHR analysis is time-consuming and often difficult to interpret, with a low sensitivity and a resulting high false-positive rate.[3] In addition, FHR changes may only represent current cardiorespiratory status of the fetoplacental unit and not reflect chronic placental reserves.

Table I. Baseline characteristics, by study group

Characteristic	Group		P value
	Doppler	NST	
No.	649	691	NS
Maternal age (%)			
≤30 y	35	35	NS
30-35 y	36	36	NS
Gravidity (%)			
Primigravid	29	29	NS
Multigravid	64	59	NS
Gestational age at entry (mo)*	39.7 ± 3.1	39.8 ± 1.7	NS
Induction for abnormal testing (No.)	31 (4.8%)	13 (1.9%)	.005
Tests (No.)*	2 (1.2)	2 (1.5)	NS
Patients on whom amniotic fluid index was performed (No.)	111 (17%)	31 (4%)	.001
Indication for assessment (No.)			
Postdates	320 (49%)	315 (45%)	NS
Decreased fetal movement	147 (23%)	162 (23%)	NS
Diabetes mellitus	74 (11%)	76 (11%)	NS
Suspected IUGR	41 (6%)	57 (8%)	NS
Hypertension	67 (10%)	81 (12%)	NS

NS, Not significant.
*Data are presented as mean ± SD.

The blood flow patterns in the fetal umbilical artery can be measured with Doppler ultrasound techniques. Various Doppler indices, which include the systolic/diastolic ratio, can be derived from an evaluation of the fetal umbilical circulation. Measurement of umbilical artery blood flow chronicles placental perfusion. Absent or reversed blood flow in the umbilical artery during diastole reflects abnormalities in placental resistance and is associated with up to a 50% incidence of fetal compromise at birth.[4] Artery blood flow patterns are reported to precede abnormal FHR patterns by a median of 7 days.[5] Fetal umbilical artery Doppler studies provide the opportunity to identify fetal compromise earlier and thus offer the possibility for intervention before significant compromise takes place. This may allow fetuses to tolerate the stress of labor while some degree of placental reserve is still present, which decreases the need for cesarean deliveries in labor because of fetal distress.

Chronic placental insufficiency that is associated with high-risk obstetric conditions can cause fetal compromise during either the antepartum or intrapartum periods. Identification of fetal compromise may require intervention through the induction of labor or cesarean delivery. Appropriately timed intervention may reduce the risk of perinatal morbidity, admission to the neonatal intensive care unit, and long-term sequelae of asphyxia or even fetal death.[4]

This study was undertaken to determine the best screening protocol for pregnant women at a gestational age of >32 weeks whose pregnancies were at an increased risk for poor perinatal outcome because of uteroplacental compromise. The hypothesis was that the use of a protocol that involves umbilical artery Doppler studies would result in a decreased cesarean delivery rate for fetal distress in labor when compared with the nonstress test. The primary outcome measure was the incidence of cesarean delivery for fetal distress in labor.

Methods

This trial was approved by the ethics committee of the University of British Columbia; the women gave written informed consent before enrollment in the study.

Pregnant women at a gestational age of ≥32 weeks were considered eligible for the study if, during antepartum testing, any one of the following the indications were found: (1) maternal hypertensive disorders (chronic hypertension and preeclampsia), (2) diabetes mellitus that required insulin, (3) suspected intrauterine growth restriction (IUGR; fetal abdominal circumference, <10th percentile for gestational age[6]), (4) postdates (a pregnancy entering week 41, with gestational age determined by second-trimester ultrasound scanning at 16 to 20 weeks or certain menstrual dates), and (5) patient-perceived decrease in fetal movement for >24 hours.

The exclusion criteria were (1) premature rupture of membranes, (2) multiple pregnancies, (3) fetal death in utero, (4) known lethal fetal anomaly, (5) known fetal cardiovascular anomaly, and (6) women in a subsequent pregnancy if they had participated in the study in a previous pregnancy.

On the first visit for fetal testing, patients who fit the criteria were approached by one of the fetal monitoring nurses or sonographers who worked in the fetal assessment unit and who explained the study and obtained written consent for patient participation.

Table II. Intrapartum events, by study group

Characteristic	*Group (No.)*		*P value*
	Doppler	*NST*	
No.	649	691	NS
1-min Apgar score <4	32 (4.9%)	40 (5.8%)	NS
5-min Apgar score ≤7	19 (2.9%	24 (3.5%)	NS
Vaginal operative delivery	112 (17%)	117 (16.9%)	NS
Cesarean delivery fetal distress	30 (4.6%)	60 (8.7%)	<.004
Cesarean delivery total exclusive of fetal distress	153 (23.6%)	163 (23.6%)	NS
Stillbirth	0	1	NS
Meconium at birth	124 (19%)	157 (22.8%)	NS
Admission to neonatal intensive care unit	16 (2.5%)	23 (3.3%)	NS
Birth weight (g)*	3572 ± 552	3530 ± 635	NS

NS, Not significant.
*Data are given as mean ± SD.

Consenting patients were assigned randomly to undergo fetal assessment by one of two screening protocols. Randomization was performed by opening sequentially numbered opaque envelopes. Each envelope contained an allocation that had been determined with the use of a random number table with a variable block size of 4 and 6. The first protocol consisted of electronic FHR monitoring with NSTs. The second protocol used umbilical artery Doppler studies. The nurse/sonographer who obtained the patient's consent approached the unit clerk to obtain the next envelope in sequence. Envelopes were kept in a locked drawer that was accessible only to the unit clerk. The envelope was opened by the nurse/sonographer in the presence of the patient. Once assigned randomly to a particular group, the patient remained in that group for any subsequent assessments that took place in that pregnancy.

The primary outcome measure that was used to determine the sample size was the incidence of cesarean delivery for a nonreassuring FHR tracing that was indicative of fetal distress as a proxy for fetal and maternal morbidity that was created by operative delivery. A diagnosis of fetal distress was indicated in the presence of a nonreassuring FHR pattern on the intrapartum FHR tracing that was identified to be a combination of repetitive variable decelerations with decreased variability, late decelerations with or without decreased variability, or a fetal bradycardia with no recovery. Secondary outcome measures were total cesarean delivery rate, Apgar scores at 1 and 5 minutes, the incidence of stillbirth, the presence of meconium, and the incidence of transfer to the neonatal intensive care unit with severe neonatal morbidity.

Patients who were assigned randomly to group A underwent an NST that was performed by an appropriately trained registered nurse using an FHR monitor (Hewlett Packard, 8040A, Hewlett Packard, Palo Alto, Calif). The duration of the FHR monitoring was 20 to 60 minutes. Vibroacoustic stimulation was not used in this study. The NST was then interpreted by one of the maternal-fetal medicine physicians, according to standard criteria. The frequency of testing was that of the usual assessment protocol of twice per week, if the NST was reactive with 2 accelerations within 20 minutes and no decelerations. If the NST was equivocal (a Kubli score of 6)[7,8] showing normal variability but with decelerations or bradycardia, an assessment of the amniotic fluid was performed by the amniotic fluid index. The Kubli score has five separate FHR components: baseline rate, amplitude, frequency, decelerations, and accelerations. Each component is scored on a scale of 0 to 2. A maximum score of 10 is given for a reactive tracing. In the NST group if the monitoring results were abnormal (a Kubli score of <4) with decreased variability and/or the presence of FHR decelerations or prolonged tachycardia, the recommendation was made to the attending physician to proceed with induction/delivery within 24 hours.

Patients who were assigned randomly to group B underwent umbilical artery Doppler velocimetry that was performed by an appropriately trained registered nurse or sonographer. Doppler velocimetry was performed with a continuous wave or pulsed Doppler probe, which operated at 4 MHz. Studies were performed in the absence of fetal breathing and included at least six to eight continuous cardiac cycles for the determination of mean systolic peaks and diastolic troughs to obtain a systolic/diastolic ratio. The output needed to show both arterial and venous blood flow to ensure measurement of the blood flow velocity near the mid point of the artery. Each velocimetry study consisted of three separate insonations, which were averaged to achieve a final systolic/diastolic ratio. The systolic/diastolic ratios were then categorized after a comparison with an institutional nomogram of umbilical artery Doppler percentiles for gestational ages between 28 and 42 weeks.[9] A test was considered equivocal when the systolic/diastolic ratio was above the 90th percentile for gestational age. A test was considered abnormal when absent or reversed end-diastolic blood flow was detected.

When the Doppler assessment was normal, the frequency of repeat testing was twice per week. When the

Doppler assessment was equivocal (systolic/diastolic ratio, ≥90%), an assessment of amniotic fluid by the amniotic fluid index was done. When the Doppler assessment was abnormal (absent or reverse end diastolic blood flow), recommendations were made to the attending physician to proceed with induction/delivery within 24 hours.

When either the NST or Doppler assessment was equivocal, an additional assessment test that used the amniotic fluid index was performed. The combination of an equivocal NST or Doppler assessment and a normal amniotic fluid index resulted in a repeat assessment within 24 hours. The combination of an equivocal NST or Doppler assessment and the presence of oligohydramnios (an amniotic fluid index of <5th percentile for gestational age)[10] on the amniotic fluid assessment resulted in a recommendation to the attending physician to proceed to induction/delivery within 24 hours. Because of the nature of the test, neither the patients nor the physicians could be blinded about the intervention.

Sample size. A previous pilot study that was performed at our institution demonstrated that the incidence of cesarean delivery for fetal distress in the high-risk population was approximately 5.5%. To achieve a power of 80% to detect a 50% reduction in the rate of cesarean delivery with a one-sided type I error of 5%, it was calculated that a sample size of 1240 patients would be required. A one-sided α was chosen because we were interested only in determining whether Doppler assessment was better than NST as a tool for fetal assessment. This was a reasonable goal that was based on the previous studies that had been performed.

Statistical analysis. Analysis was carried out on an intention to treat basis. All women for whom outcome data were available were included in the analysis. Categoric variables were analyzed with the use of χ^2 statistics. The Fisher exact test was used where any cell numbers were <5. Continuous variables with a normal distribution were analyzed with a one-tailed unpaired *t* test. For all analyses, a probability value of <.05 was used to define statistical significance.

Results

Patients were recruited between October 1997 and October 2000, and follow-up examinations continued until all patients were delivered. During the study period, 1360 women were enrolled into the study; 95% of the women who were approached with the identified high-risk conditions agreed to participate in the study. Four patients were assigned randomly in error and did not have the identified high-risk condition and were removed from further analysis. There was randomization compliance with the monitoring protocol in 1356 women. Of the women who were enrolled in the study, no final outcome data were available for 16 patients. Of the 16 patients who were lost to follow up, 10 patients were in the NST group, and 6 patients were in the Doppler group. Analysis occurred for 1340 women (Doppler group, 649 women; NST group, 691 women). There were no significant differences that were identified in the maternal demographic data between the NST and the Doppler groups (Table I). There was a higher incidence of induction in the Doppler group for abnormal testing (31 women [4.8%] vs 13 women [1.9%]; $P < .005$). For the 1340 women, the rate of cesarean delivery for indicated fetal distress was significantly higher in the NST group (60 women [8.7%]) versus the Doppler group (30 women [4.6%]; $P \leq .006$; Table II). The overall cesarean delivery rate and the specific indications for cesarean delivery were otherwise similar between the groups, the only exception being for the indication of fetal distress (Table III). When the cesarean delivery rate for fetal distress was explored on the basis of the indication for testing, the highest reduction between the two testing arms was for the indications of maternal hypertension, suspected IUGR, diabetes mellitus, and decreased fetal movement (Table IV). There were no significant differences in Apgar scores at 1 minute and 5 minutes, in operative vaginal delivery, in meconium staining, and in admissions to the neonatal intensive care unit between the two study groups. There was one stillbirth in the NST group.

Table III. Indication for cesarean delivery, by study group

	Group (No.)		
Indication	*Doppler*	*NST*	*P value*
Total cesarean deliveries	183	223	
Malpresentation: breech	18 (10%)	20 (9%)	NS
Failure to progress	98 (54%)	108 (48%)	NS
Fetal distress	30 (16%)	60 (27%)	.011
Planned repeat	26 (14%)	23 (10%)	NS
Miscellaneous (bleeding)	11 (6%)	12 (5%)	NS

NS, Not significant.

Comment

Kubli et al[7] and Lyons et al[8] proposed the usefulness of the evaluation of FHR patterns in the absence of contractions. Fetal hypoxia and/or acidosis is associated with abnormalities in the central control of the FHR that are expressed through the autonomic nervous system. This results in changes in the variability of the FHR pattern, in the baseline FHR, and in the presence of FHR decelerations.[2]

The use of NSTs for antenatal testing in several large studies show no improvement in perinatal outcome, with a sensitivity to predict perinatal morbidity of 60% and a specificity of <50%.[11-14] In the Cochrane meta-analysis of four trials in which electronic fetal monitoring was the primary means of surveillance, there was no significant

Table IV. Incidence of cesarean delivery for fetal distress by test indication

Indication	Group (No.)		Change in incidence (%)
	Doppler	NST	
Total cesarean deliveries for fetal distress	30	60	20
Postdates	18 (5.6%)	21 (6.7%)	
Decreased fetal movement	7 (4.8%)	14 (8.6%)	79
Diabetes mellitus	3 (4%)	9 (11.8%)	195
Hypertension	1 (1.5%)	11 (13.6%)	807
Suspected IUGR	1 (2.4%)	5 (8.8%)	267

improvement in the rate of admissions to the neonatal intensive care unit (20.8% vs 19.8%),[15] cesarean delivery rates, the incidence of low Apgar scores at birth, or abnormal neonatal neurologic signs.

Doppler ultrasound scanning was introduced to investigate the umbilical artery blood flow pattern in high-risk pregnancies that were at risk for fetal compromise.[16] Some high-risk pregnancies, notably those complicated by small for gestational age, are associated with a high placental resistance, which reflects a major defect in fetoplacental perfusion.[17] The result from a recent meta-analysis shows that, in high-risk pregnancies that are treated with a similar Doppler protocol, there was a significant decrease in the perinatal mortality rate when lethal anomalies were excluded.[18] In pregnancies that were complicated by hypertension and suspected IUGR, umbilical Doppler ultrasound scanning was associated with a 29% reduction in the odds of perinatal death.[18]

The results of the present study show that the use of umbilical artery Doppler ultrasound assessment as an antenatal screening test is associated with a reduction in the incidence of cesarean delivery for fetal distress. This study had one stillbirth in the NST group, but no deaths in the Doppler group, which was a difference that was likely due to chance. The sample size was not sufficient to detect a significant reduction in perinatal death, because an estimated sample size of 19,000 women would be required. Although our study took place over a 3-year period, there was no real change in the patterns of practice in terms of delivery for fetal distress.

More patients in the Doppler group required an induction of labor than in NST group, which was consistent with our belief that Doppler assessment identified a higher proportion of patients with early placental compromise than the NST assessment. This increased rate of induction did not result in an increase in the overall rate of cesarean delivery. In addition, there was no increase in the rate of cesarean delivery for failure to progress in the Doppler group.

Obviously, the decision to proceed with a cesarean delivery is a complex decision, with many factors identified in the literature that are difficult to quantify. Even the use of protocols does not always help to standardize the process. In a randomized trial, these factors should affect both groups equally, unless there is something intrinsic in the way the attending physicians respond to the information that is provided by one test over the other. It was not possible to blind either the patients or their physicians as to the protocol to which the patient had been allocated. This could introduce a potential source of bias, if knowledge of the testing allocation influenced the attending physician's decision-making during the course of labor and delivery. We were very careful to give the attending physicians the same recommendation for treatment when there was abnormal testing in both groups. The standard recommendation was for delivery within 24 hours. No cases occurred for which a recommendation for induction was bypassed and a primary cesarean delivery was performed. If a bias had been introduced, we would have expected to see a difference in the overall cesarean delivery rate rather than a difference appearing only for the indication of fetal distress in labor.

Our study was designed to detect a reduction in the cesarean delivery rate in the study population as a whole. There are inherent difficulties in subgroup analysis in the absence of sufficient power. Valid and reliable conclusions cannot be made from such information. As a result, we decided not to perform statistical analysis on the data that involved differences in the cesarean rate for fetal distress between the various subgroups that were created by the testing indication. Table IV does highlight that there may be greater benefit to Doppler evaluation in cases that involve hypertension or suspected IUGR, because the greatest difference in the cesarean delivery rate occurred in these 2 subgroups. This would require further study for definitive conclusions to be drawn. This study is in agreement with other literature reports that show a lack of improvement in outcomes when umbilical Doppler ultrasound scanning is used in postdate gestations.[19,20]

The likely explanation for our findings of a trend toward reduced cesarean delivery for fetal distress risk in patients in the hypertensive and suspected IUGR and diabetic groups is predicated by our understanding of the high resistance to blood flow that is documented in the umbilical artery. The lack of a similar beneficial effect in the postdate group and only a moderate trend in the fetal movement groups suggests that the mechanism for poor fetal outcome in these groups is not mediated completely by a chronic decrease in placental reserve that is capable of being assessed appropriately by umbilical artery Doppler scanning.

The current study concludes that umbilical artery Doppler ultrasound assessment of the fetus in conjunction with a policy of directed intervention results in a decrease in the rate of cesarean delivery for fetal distress in a high-risk population, when compared with the NST. The introduction Doppler ultrasound assessment as a primary screening test must be balanced against a higher rate of secondary testing by amniotic fluid index, training of personnel to perform these tests, and the purchase of new machinery.

Larger prospective randomized, controlled trials are necessary to determine the significance of this testing protocol in specific subgroups for the reduction in other aspects of perinatal morbidity and death.

REFERENCES

1. Platt LD, Paul RH, Phelan J, Walla CA, Broussard P. Fifteen years of experience with antepartum fetal testing. Am J Obstet Gynecol 1987;156:1509-15.
2. Dawes GS, Harkness RA, Howlett TA, et al. Antenatal fetal heart variation in relation to the respiratory and metabolic status of the compromised human fetus. Br J Obstet Gynaecol 1988;95:980-9.
3. Thacker SB, Berkelman RL. Assessing the diagnostic accuracy and efficacy of selected antepartum fetal surveillance techniques. Obstet Gynecol Surv 1986;41:121-41.
4. Karsdorp VHM, Vugt van JMG, Geijn van HP, Kostense PJ, Arduini D, Montenegro N, et al. Clinical significance of absent or reversed end diastolic velocity waveforms in umbilical artery. Lancet 1994;344:1664-8.
5. Arduini D, Rizzo G, Romanini C. The development of abnormal heart rate patterns after absent end-diastolic velocity in umbilical: analysis of risk factors. Am J Obstet Gynecol 1993;168:43-50.
6. Lessoway VA, Schulzer M, Wittman BK, Gagnon FA, Wilson RD. Ultrasound fetal biometry charts for a North American Caucasian population. J Clin Ultrasound 1998;26:433-53.
7. Kubli F, Boos R, Ruttgers H, Hagens CV, Vanselow H. Antepartum fetal heart rate monitoring. In: Beard RW, Campbell S, editors. Current status of fetal heart rate monitoring and ultrasound in obstetrics. London: Royal College of Obstetricians and Gynecologists; 1977. p. 28-45.
8. Lyons ER, Ylsma-Howell M, Shamsi S, Towel ME. A scoring system for non-stressed antepartum fetal heart rate monitoring. Am J Obstet Gynecol 1979;133:242-6.
9. Lopes LM, Bebbington M, Lessoway V, Farquharson DF, Dansereau J, Shaw D. Doppler fluxometria da arteriaumbilical: geraccao de una curva normalidad. Pontos Obstet Gincol 1993;1:80-4.
10. Moore TR, Cayle JE. The amniotic fluid index in normal pregnancy. Am J Obstet Gynecol 1990;162:1168-73.
11. Schneider EP, Hutson JM, Petrie RH. An assessment of the first decade's experience with antepartum fetal heart rate testing. Am J Perinatol 1988;5:134-41.
12. Phelan JP. The nonstress test: a review of 3,000 tests. Am J Obstet Gynecol 1981;····:7-10.
13. Kidd LC, Patel NB, Smith R. Non stress antenatal cardiotocography: a prospective randomized controlled trial. Br J Obstet Gynaecol 1985;92:1156-9.
14. Brown VA, Sawers RS, Parson RJ, Duncan SLB, Cooke ID. The value of antenatal cardiotocography in the management of high-risk pregnancy: a randomized controlled trial. Br J Obstet Gynaecol 1982;89:716-22.
15. Neilson JP, Cardiotocography for antepartum fetal assessment [revised 12 May 1994]. In: Keirse MJNC, Renfrew MJ, Neilson JP, Crowther C, editors. Pregnancy and childbirth module: Cochrane Pregnancy and Childbirth Database; Cochrane Collaboration, Issue 2. Oxford: Update Software; 1994.
16. Fitzgerald DE, Drumm JE. Non-invasive measurements of human fetal circulation using ultrasound: a new method. BMJ 1977;2:1450-1.
17. Divon MY, Griz BA, Lieeblich R, Langer O. Clinical management of the fetus with markedly diminished umbilical artery end-diastolic flow. Am J Obstet Gynecol 1989;161:1523-7.
18. Neilson JP, Alfirevicz A. Doppler ultrasound in high-risk pregnancies [revised 23 July 1995]. In: Keirse MJNC, Renfrew MJ, Neilson JP, Crowther C, editors. Pregnancy and childbirth module: Cochrane Pregnancy and Childbirth Database. Cochrane Collaboration, Issue 2. Oxford: Update Software; 1995.
19. Weiner Z, Farmakides G, Schulman H, Kellner L, Plancher S, Maulik D. Computerized analysis of fetal heart rate variation in post term pregnancy: prediction of intrapartum fetal distress and fetal acidosis. Am J Obstet Gynecol 1994;171:1132-8.
20. Battaglia C, Larocca E, Lanzani A, Coukos G, Genazzani A. Doppler velocimetry in prolonged pregnancy. Obstet Gynecol 1994;77:213-6.